

2020 AMC 10A Problem 16 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 16. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 16

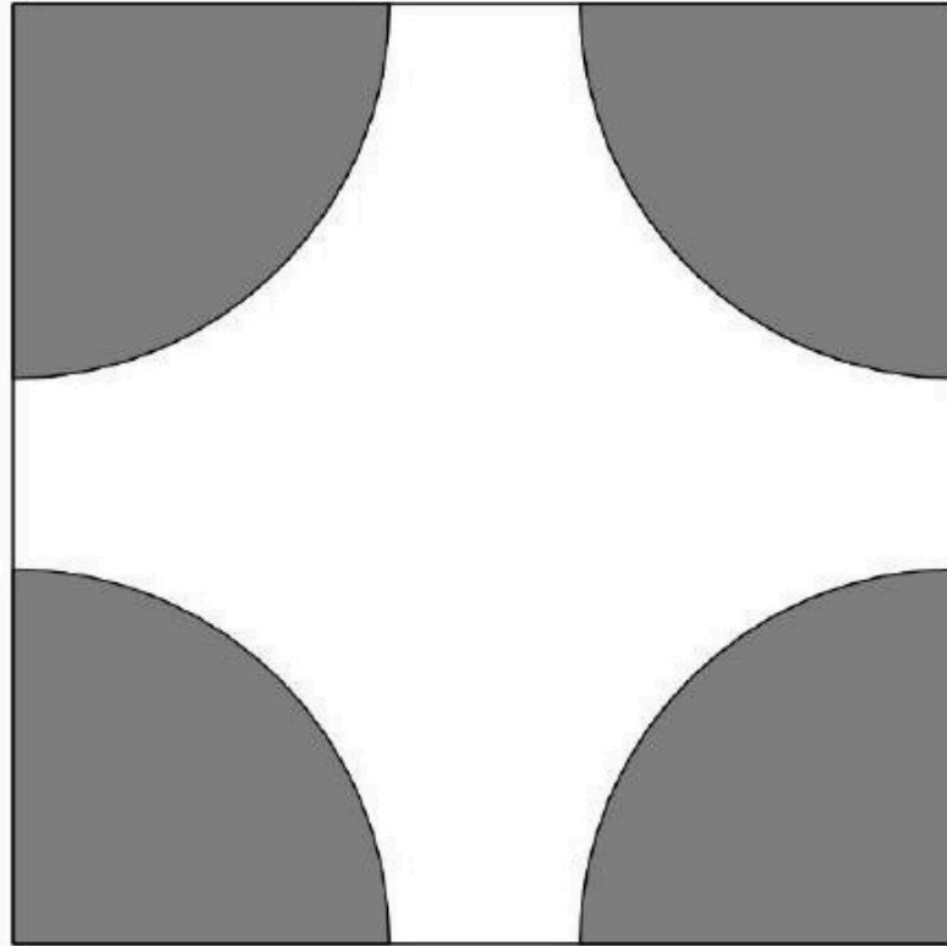
Problem:

A point is chosen at random within the square in the coordinate plane whose vertices are $(0, 0)$, $(2020, 0)$, $(2020, 2020)$, and $(0, 2020)$. The probability that the point lies within d units of a lattice point is $\frac{1}{2}$. (A point (x, y) is a lattice point if x and y are both integers.) What is d to the nearest tenth?

Answer Choices:

- A. 0.3
- B. 0.4
- C. 0.5
- D. 0.6
- E. 0.7

Solution:



The diagram represents each unit square of the given 2020×2020 square.

We consider an individual one-by-one block. If we draw a quarter of a circle from each corner (where the lattice points are located), each with radius d , the area covered by the circles should be 0.5. Because of this, and the fact that there are four circles, we write

$$4 \cdot \frac{1}{4} \cdot \pi d^2 = \frac{1}{2}$$

Solving for d , we obtain

$$d = \frac{1}{\sqrt{2\pi}}, \text{ where with } \pi \approx 3, \text{ we get } d \approx \frac{1}{\sqrt{6}} \approx \frac{1}{2.5} = \frac{10}{25} = \frac{2}{5}$$

and from here, we see that

$$d \approx \boxed{(B) 0.4}$$

Note: To be more rigorous, note that $d < 0.5$ since if $d \geq 0.5$ then clearly the probability is greater than $\frac{1}{2}$. This would make sure the above solution works as if $d \geq 0.5$ there is overlap with the quarter circles.

OR

As in the previous solution, we obtain the equation

$$4 \cdot \frac{1}{4} \cdot \pi d^2 = \frac{1}{2}$$

which simplifies to

$$\pi d^2 = \frac{1}{2} = 0.5$$

Since π is slightly more than 3, d^2 is slightly less than $\frac{0.5}{3} = 0.1\bar{6}$. We notice that $0.1\bar{6}$ is slightly more than $0.4^2 = 0.16$, so

$$d \text{ is roughly } \boxed{(B) 0.4}$$

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